THE CYLINDER

SOLID MENSURATION

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

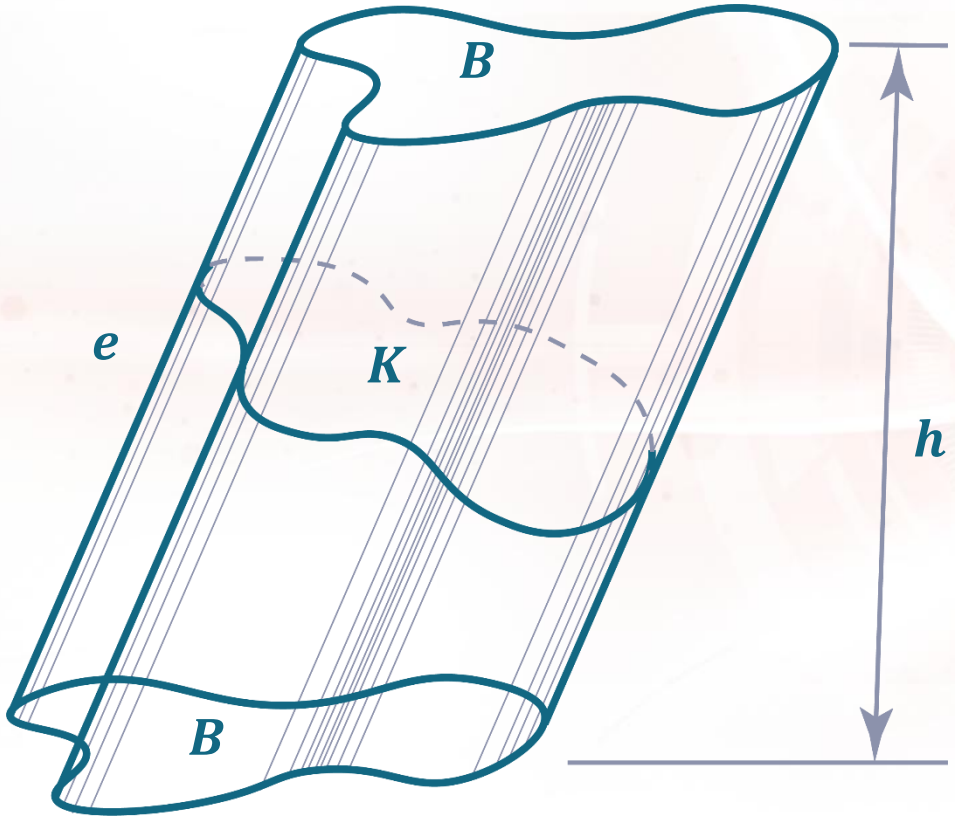
TOPIC OUTLINE

Properties of a Cylinder

- Lateral Surface Area
- Total Surface Area
- Volume

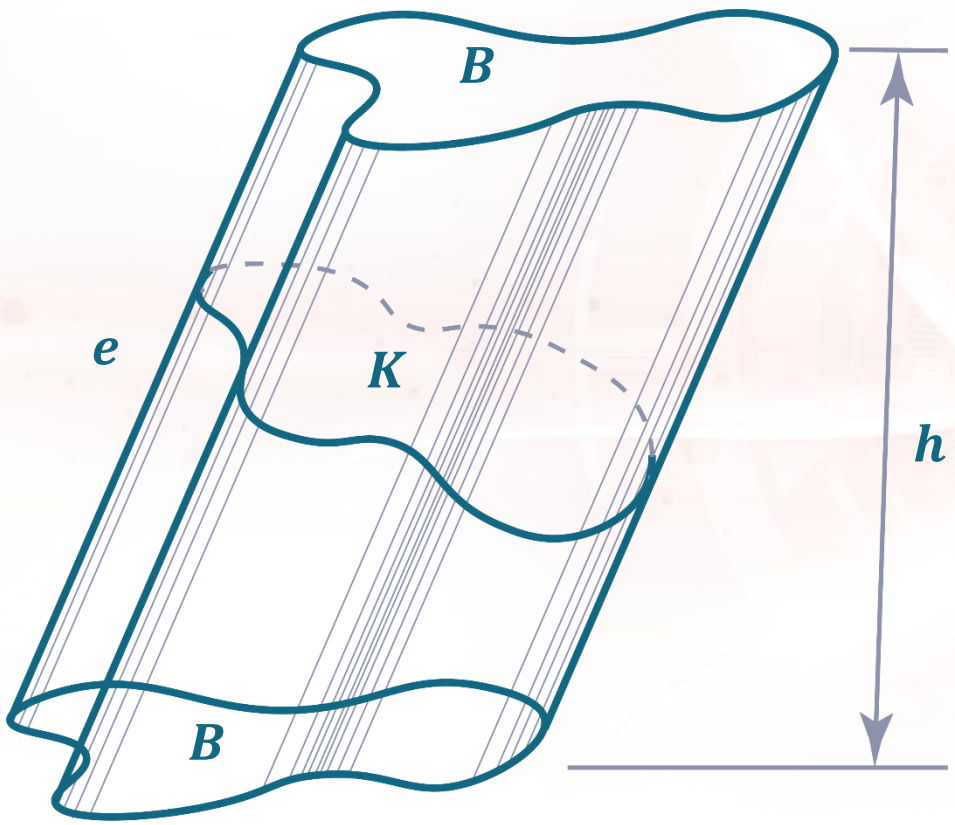
PROPERTIES OF A CYLINDER

DEFINITION



A **cylinder** is a solid bounded by a closed **cylindrical surface** and two parallel planes

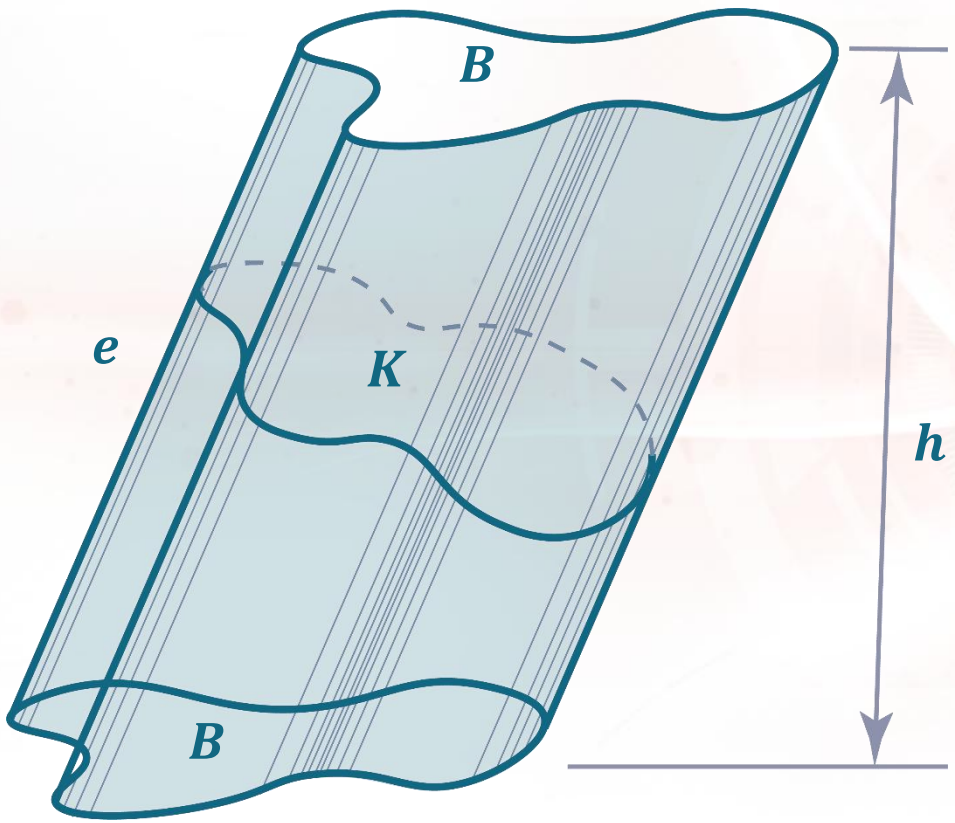
PROPERTIES



1. The **bases, *B*** of a cylinder are equal.
2. The **altitude, *h*** of a cylinder is the perpendicular distance between the bases.
3. The **right section, *K*** of a cylinder is a section perpendicular to all elements of the cylinder.
4. The **elements, *e*** of a cylinder are equal.

AREA OF A CYLINDER

LATERAL SURFACE AREA



The Lateral Surface Area **S**

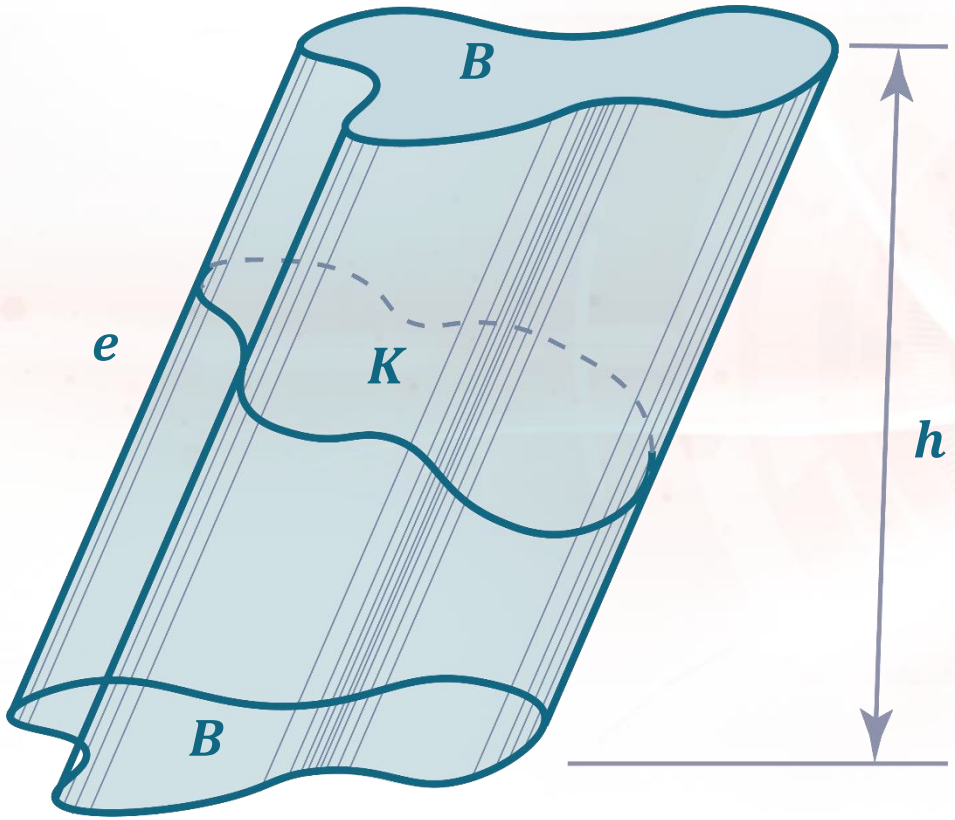
$$S = eP$$

where

e = lateral edge

P = perimeter of right section

TOTAL SURFACE AREA



The Total Surface Area **T**

$$T = 2B + S$$

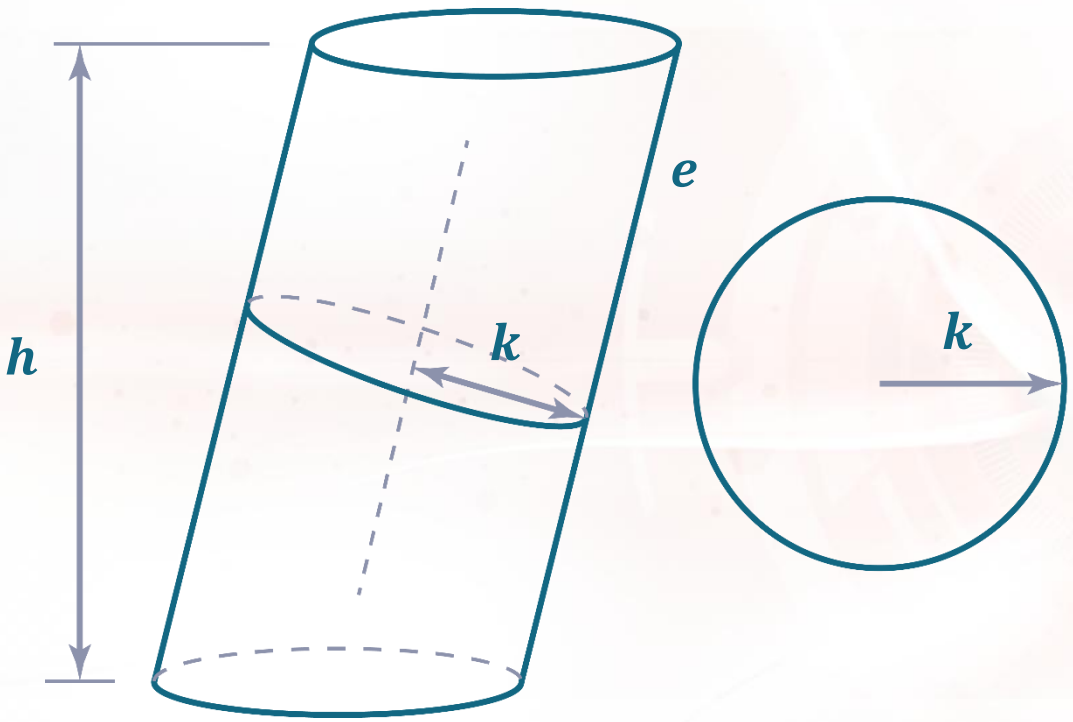
where

B = area of either base

S = lateral surface area

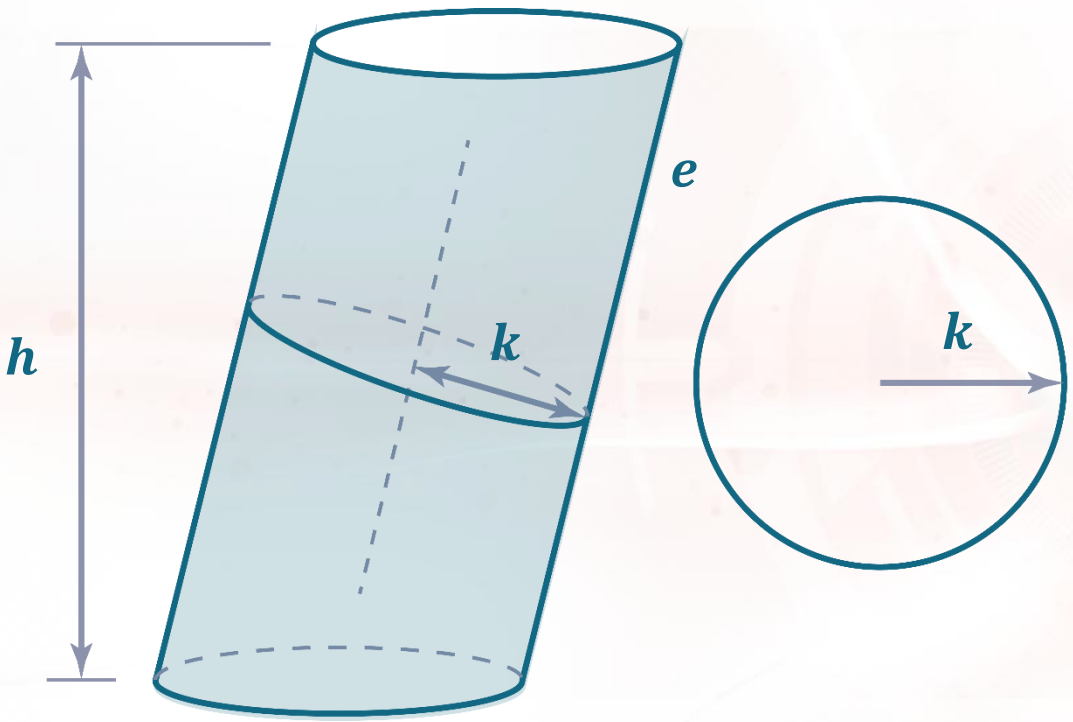
CIRCULAR CYLINDER

DEFINITION



A circular cylinder is a cylinder which has a **circular right section**

LATERAL SURFACE AREA



The Lateral Surface Area **S**

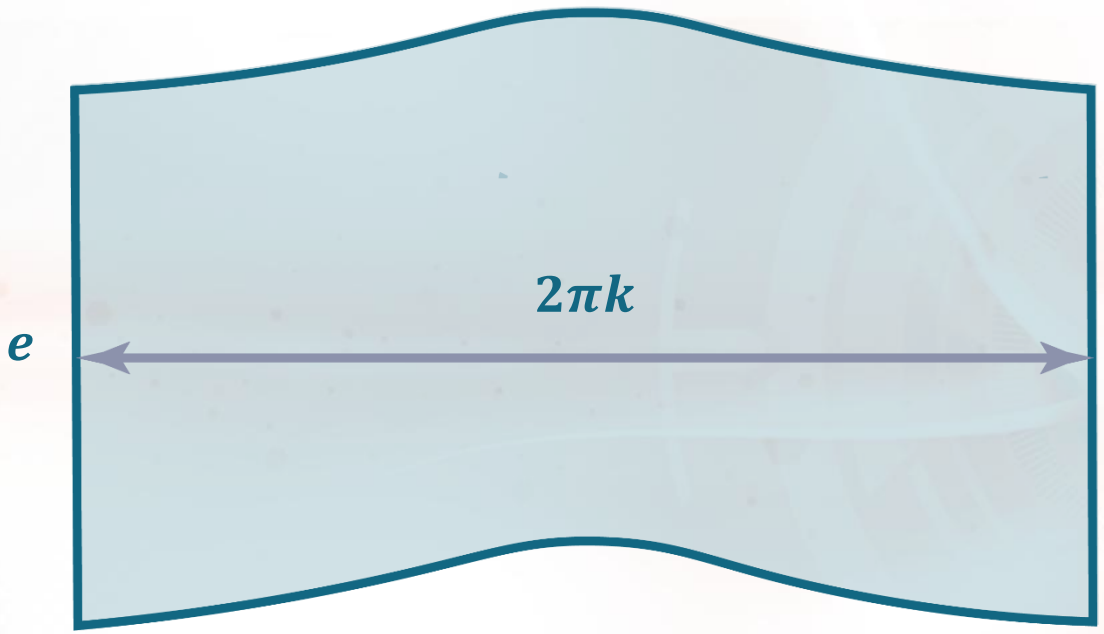
$$S = eP$$

$$\boxed{S = 2\pi ke}$$

where

k = radius of right section

$e = element$



The Lateral Surface Area **S**

$$s = eP$$

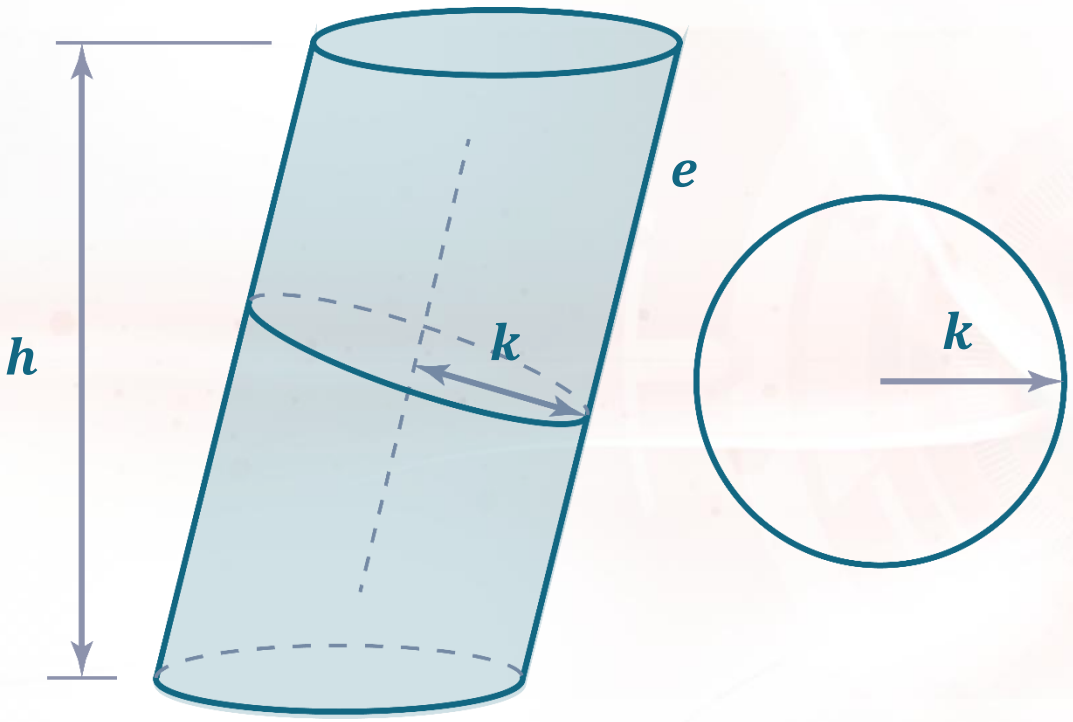
$$\boxed{S = 2\pi k e}$$

where

k = radius of right section

$e = element$

TOTAL SURFACE AREA



The Total Surface Area **T**

$$T = 2B + S$$

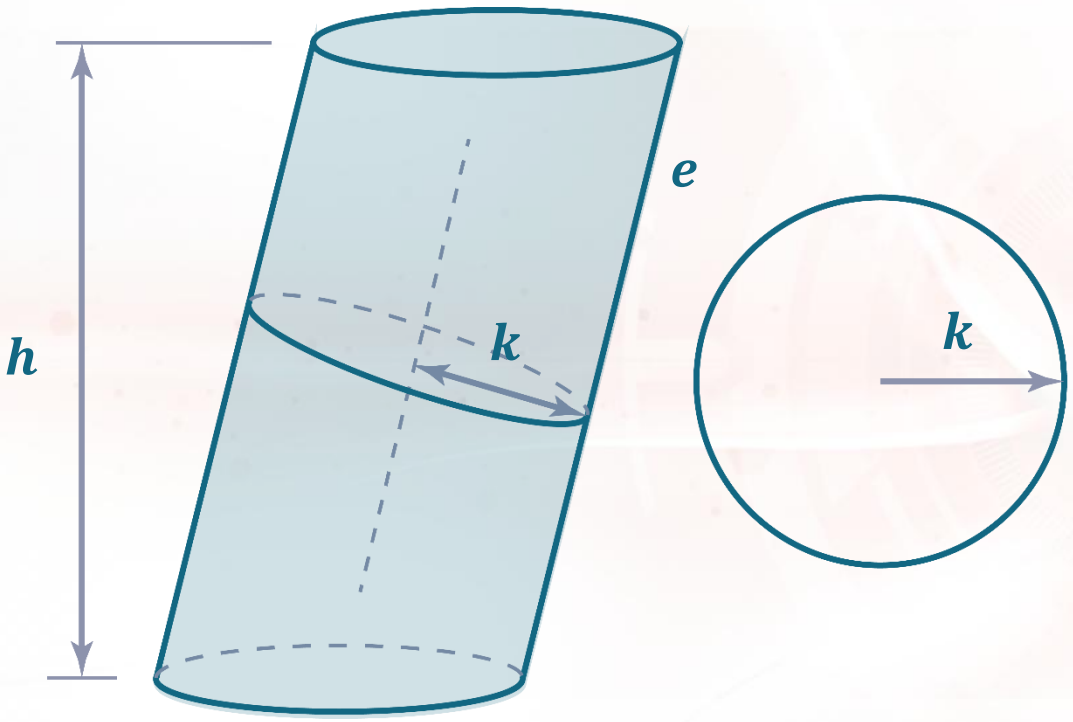
where

B = area of either base

S = lateral surface area

EXERCISE

Consider a metal **pipe** with a **radius** of **4** inches and a **height** of **10** inches. Calculate the **cost** of metal required to make it, assuming that the metal costs **$0.05 per square inch**.



The Volume **V**

$$V = Bh$$

$$\boxed{V = \pi k^2 e}$$

where

$k = radius\ of\ right\ section$

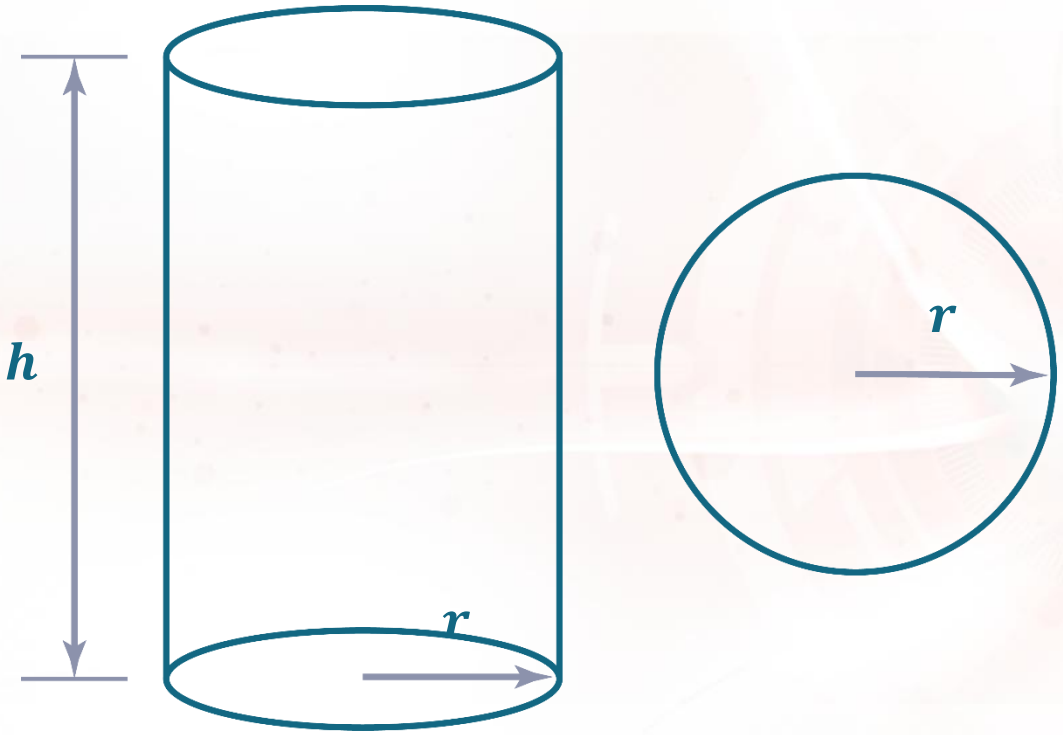
$e = element$

EXERCISE

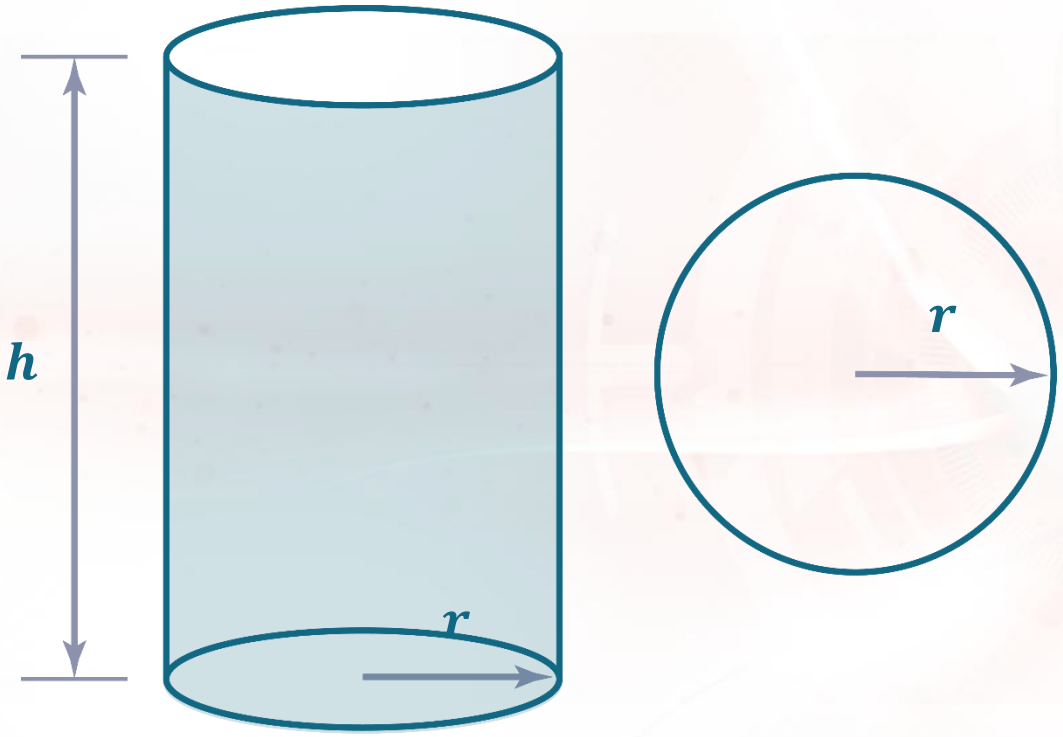
Consider a **cylindrical** container that is **12** inches **tall** and has a **radius** of **6** inches. Calculate the **volume** of the container, as well as the **cost** of filling it with water, assuming that the cost of water is **$0.01 per cubic inch**.

RIGHT CIRCULAR CYLINDER

DEFINITION



A **right** circular cylinder is a cylinder whose elements are **perpendicular** to its base and whose bases are circles



The Lateral Surface Area **S**

$$S = eP$$

$$\boxed{S = 2\pi rh}$$

where

r = radius of base

$h = altitude$

The Lateral Surface Area **S**

$$S = eP$$

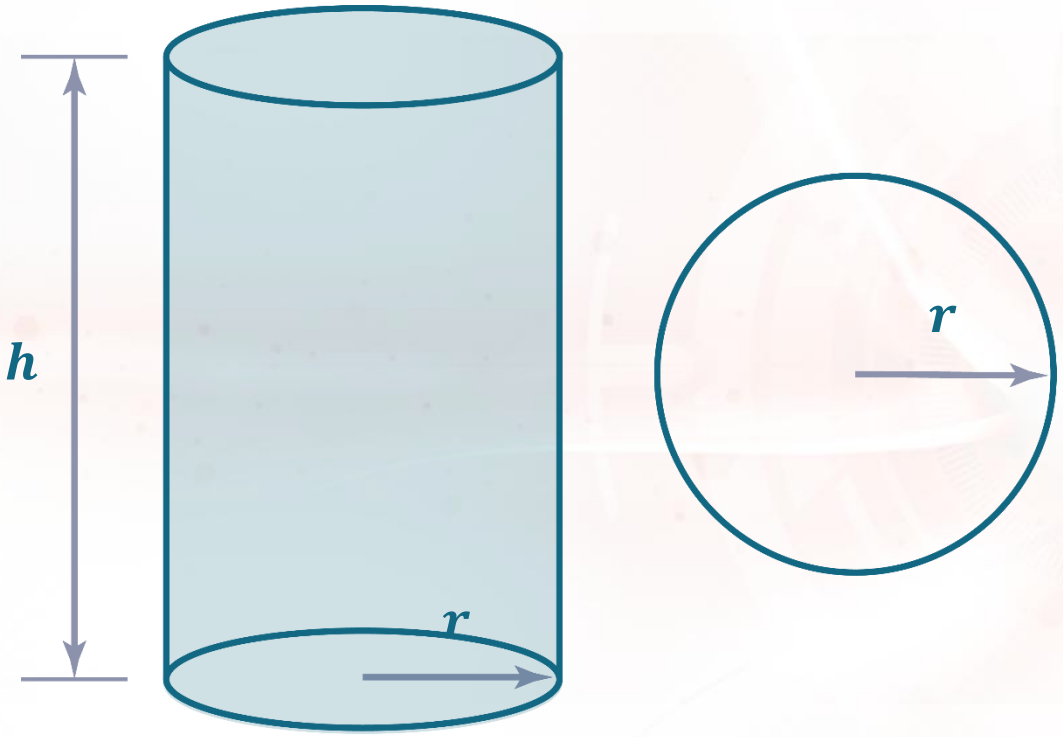
$$\boxed{S = 2\pi rh}$$

where

r = radius of base

$h = altitude$

TOTAL SURFACE AREA



The Total Surface Area **T**

$$T = 2B + S$$

$$\boxed{T = 2\pi r(h + r)}$$

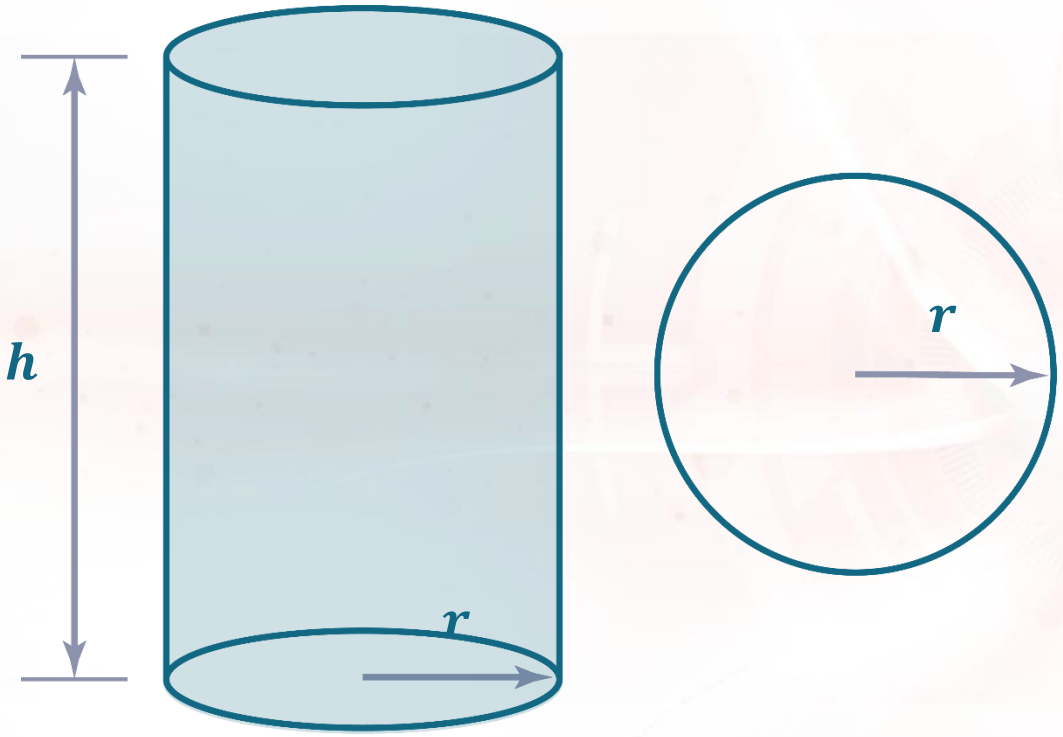
where

r = radius of base

h = altitude

EXERCISE

Consider a **cylindrical** pipe with a **diameter** of **8** inches and a **height** of **24** inches. The pipe has a cylindrical **hole** drilled straight through the center with a **diameter** of **4** inches that runs the entire height of the pipe. Calculate the **cost** of painting the pipe assuming the cost of paint is **$0.10 per sq. in**.



The Volume **V**

$$V = Bh$$

$$\boxed{V = \pi r^2 h}$$

where

r = radius of base

h = altitude

Consider a hollow **right circular cylinder** with an **outer** radius of **10** cm, an **inner** radius of **8** cm, and a **height** of **20** cm. Calculate the **cost** of metal used to make it, assuming that the cost of the metal is **$0.05 per cubic centimeter**.

SEATWORK